

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses only the Linear Regression algorithm	Prediction accuracy may be limited for complex datasets	High
2	Supports prediction for a single user at a time	Reduced performance under high user traffic	Medium
3	Limited dataset size	May affect prediction accuracy and generalization	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on scalable cloud infrastructure with load balancing
2	Data Storage	Uses a CSV dataset	Integrate a relational database such as MySQL or PostgreSQL
3	Performance	Single Linear Regression model	Optimize the model and implement caching for faster predictions
4	Security	Basic Flask application	Add user authentication, HTTPS, and input validation

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Integrate advanced machine learning models such as Random Forest and XGBoost	2 Months	Improve prediction accuracy
Phase 3	Add interactive dashboards, charts, and country-wise HDI comparisons	5 Months	Better data visualization and user experience
Phase 4	Deploy a mobile-friendly application with user authentication and prediction history	9 Months	Improved accessibility, security, and scalability